

基于ARM+FPGA异构平台的目标检测加速模块设计与实现

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摘要 为解决基于深度学习目标检测模型规模大、在边缘设备上难以部署的问题, 以YOLO目标检测模型为例, 设计实现基于ARM+FPGA异构平台的目标检测加速模块。该系统使用剪枝、量化后的压缩模型, 在FPGA实现神经网络前向推理加速, 在ARM中实现加速器调度。实验结果表明, 部署至Xilinx ZCU102开发板上, 该模块在200 MHz工作频率下, 平均计算性能达到425.8 GOP/s, 推理压缩模型速度达到30.3 fps, 模块功耗为3.56 W, 证明该加速模块具备可配置性。

关键词 深度学习; 目标检测; 模型剪枝量化; 异构平台; 边缘计算

Design and Implementation of Object Detection Acceleration Module Based on an ARM+FPGA Heterogeneous Platform

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Abstract Object detection algorithms based on deep learning use big models are difficult to be deployed at the edge. Taking YOLO (you only look once) object detection algorithm as an example, an acceleration module based on an ARM+FPGA heterogeneous platform is proposed. The FPGA chip accelerates the forward process of the compressed model while ARM is responsible for process scheduling. Experiment results show that the peak performance of the system reaches 425.8 GOP/s under 200 MHz working frequency. The system on a Xilinx ZCU102 board achieves a frame rate at 30.3 fps, while the power consumption is 3.56 W. It is also configurable.

Key words deep learning; object detection; model pruning and quantization; heterogeneous platform; edge computing

卷积神经网络(convolutional neural network, CNN)广泛应用于目标检测和图像分类。由于CNN的参数多、计算量大, 在处理这类任务时通用处理器效率较低。因此, 基于GPU(graphics processing unit), FPGA(field programmable gate array)和ASIC(application specific integrated circuit)的专用神经网络加速器已经成为研究热点^[1]。在这些方法中, 基

于FPGA的加速器获得更多的关注。与GPU相比, FPGA的能效比更高; 与ASIC相比, FPGA的动态可重构性能适应深度学习算法的不断变化^[2-3]。

在目标检测领域, YOLO(you only look once)具有检测速度快、网络较为简单的特点, 适合部署在端侧平台^[4]。随着算法的更新, YOLO的后续版本使用更深的网络来实现更高的检测精度。YOLOv3